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AIRCRAFT ACOUSTIC PANEL WITH INTEGRATED FIRE-RESISTANT MATERIAL

Abstract

An apparatus is provided for an aircraft. This apparatus includes an acoustic panel, and the acoustic panel includes a face skin, a back skin and a cellular core disposed between and connected to the face skin and the back skin. The back skin laterally overlaps the face skin and includes an aerogel layer. The cellular core includes a plurality of cavities that extend through the cellular core from the face skin to the back skin. Each of the cavities is fluidly coupled with one or more perforations in the face skin.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to an aircraft and, more particularly, to a component of the aircraft with fire-resistant material.

BACKGROUND INFORMATION

[0002] An aircraft propulsion system may include one or more acoustic panels for attenuating sound generated by its gas turbine engine. Various types and configurations of acoustic panels are known in the art. Various fire blankets are also known in the art. While these known acoustic panels and fire blankets may be suitable for their intended purposes, there is still room in the art for improvement.

SUMMARY

[0003] According to an aspect of the present disclosure, an apparatus is provided for an aircraft. This apparatus includes an acoustic panel, and the acoustic panel includes a face skin, a back skin and a cellular core disposed between and connected to the face skin and the back skin. The back skin laterally overlaps the face skin and includes an aerogel layer. The cellular core includes a plurality of cavities that extend through the cellular core from the face skin to the back skin. Each of the cavities is fluidly coupled with one or more perforations in the face skin.

[0004] According to another aspect of the present disclosure, another apparatus is provided for an aircraft. This apparatus includes a first skin, a second skin and a metal core. The first skin is configured from or otherwise includes a first fiber-reinforced composite. The second skin laterally overlaps the first skin. The second skin includes a second fiber-reinforced composite and a fire-resistant material bonded to and laterally overlapping the second fiber-reinforced composite. The metal core is disposed between and bonded to the first skin and the second skin. The metal core includes a plurality of cavities that extend through the metal core from the first skin to the second skin.

[0005] According to still another aspect of the present disclosure, another apparatus is provided for an aircraft. This apparatus includes a first skin, a second skin and a cellular core. The second skin laterally overlaps the first skin and includes an aerogel layer. The cellular core is disposed between and bonded to the first skin and the second skin. The cellular core includes a plurality of cavities that extend through the cellular core from the first skin to the second skin. A thickness of the second skin changes as the aerogel layer extends laterally along the plurality of cavities.

[0006] The apparatus may also include an acoustic panel. The acoustic panel may include the first skin, the second skin and the cellular core.

[0007] Each of the cavities may be fluidly coupled with one or more apertures extending through the first skin.

[0008] The apparatus may also include a duct including the first skin, the second skin and the metal core. The first skin may form a peripheral boundary of a flowpath extending through the duct.

[0009] The fire-resistant material may be configured as or otherwise include an aerogel material.

[0010] The aerogel layer may laterally overlap each of the cavities.

[0011] The back skin may also include a fiber-reinforced layer bonded to and laterally overlapping the aerogel layer.

[0012] The fiber-reinforced layer may completely laterally overlap the aerogel layer. The aerogel layer may partially laterally overlap the fiber-reinforced layer.

[0013] The fiber-reinforced layer may be disposed between the cellular core and the aerogel layer.

[0014] The fiber-reinforced layer may be a first fiber-reinforced layer. The back skin may also include a second fiber-reinforced layer bonded to and laterally overlapping the aerogel layer. The aerogel layer may be disposed between the first fiber-reinforced layer and the second fiber-reinforced layer.

[0015] The aerogel layer may be disposed between the fiber-reinforced layer and the cellular core.

[0016] The fiber-reinforced layer may include fiber reinforcement embedded within a polymer

matrix.
[0017] The fiber reinforcement may be configured as or otherwise include fiberglass.
[0018] The fiber reinforcement may be configured as or otherwise include carbon fiber.
[0019] A thickness of the back skin may decrease as the back skin extends away from an edge of the aerogel layer into an intermediate region of the aerogel layer.
[0020] The face skin may include fiber reinforcement embedded within a polymer matrix.
[0021] The cellular core may be configured as or otherwise include metal.
[0022] The cellular core may be configured as or otherwise include a honeycomb core.
[0023] The acoustic panel may extend axially along and circumferentially about an axis. The back skin may be disposed radially outboard of the face skin.
[0024] The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
[0025] The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] FIG. **1** is a partial side sectional illustration of an acoustic panel arranged with aircraft components.
[0027] FIG. **2** is a partial cross-sectional illustration of the acoustic panel arranged with the aircraft components.
[0028] FIG. **3** is a partial sectional illustration of a perforated face skin.
[0029] FIG. **4** is a partial sectional illustration of a non-perforated back skin.
[0030] FIG. **5** is a partial perspective cutaway illustration of the acoustic panel.
[0031] FIGS. **6-9** are partial sectional illustrations of the acoustic panel with various fire-resistant layer arrangements.
[0032] FIG. **10** is a schematic side sectional illustration of an aircraft propulsion system configured with one or more of the acoustic panels.
DETAILED DESCRIPTION
[0033] FIGS. **1** and **2** illustrate a structural, acoustic panel **20** for an aircraft. This acoustic panel **20** may be configured to attenuate sound (e.g., noise) generated by a propulsion system of the aircraft. The aircraft propulsion system may be a turbofan propulsion system, a turbojet propulsion system, a turboprop propulsion system or any other ducted-rotor and/or open-rotor aircraft propulsion system. The acoustic panel **20** may be part of a housing (e.g., a nacelle) for an engine (e.g., a gas turbine engine, a rotary engine, a reciprocating piston engine, etc.) and/or an electric motor of the aircraft propulsion system. The acoustic panel **20**, for example, may be configured as or otherwise included as part of an inner barrel, an outer barrel, a translating sleeve, a blocker door, a bifurcation, etc. Alternatively, the acoustic panel **20** may be part of another component of the aircraft such as, but not limited to, an engine pylon, an aircraft wing or an aircraft fuselage. Furthermore, the acoustic panel **20** may also or alternatively be configured to attenuate aircraft related sound other than the sound generated by the aircraft propulsion system. Moreover, while the acoustic panel **20** is describe with reference to the aircraft propulsion system, the acoustic panel **20** may alternatively be arranged with an auxiliary power unit (APU) for the aircraft, or any other device which generates noise to be attenuated. However, for ease of description, the acoustic panel **20** of FIGS. **1** and **2** is described below as attenuating propulsion system sound and with respect to a component **22** (e.g., barrel) of the engine housing along a flowpath **24** (e.g., a bypass flowpath) within the aircraft propulsion system.
[0034] Referring to FIG. **1**, the acoustic panel **20** extends axially along an axis **26**. Briefly, this axis

26 may be a centerline axis of the aircraft propulsion system, a centerline axis of the engine housing and/or a centerline axis of the component **22** (e.g., the barrel) which is formed by or otherwise includes the acoustic panel **20**. The acoustic panel **20** extends radially from a radial inner side **28** of the acoustic panel **20** to a radial outer side **30** of the acoustic panel **20**. Referring to FIG. **2**, the acoustic panel **20** extends circumferentially about (e.g., partially or completely around) the axis **26**. The component **22** and/or its acoustic panel **20** may thereby have a curved (e.g., arcuate, cylindrical, conical, frustoconical) geometry.

[0035] With the arrangement of FIGS. **1** and **2**, a vertical thickness **32** of the acoustic panel **20** extends in a radial direction relative to the axis **26**, and a lateral plane of the acoustic panel **20** extends axially along and circumferentially about the axis **26**. The present disclosure, however, is not limited to such an exemplary curved geometry nor such an orientation relative to the axis **26**. For example, where the acoustic panel **20** is configured as or part of a sidewall of the bifurcation, the vertical thickness **32** may extend tangentially to a circular reference line about the axis **26**, and the lateral plane may extend axially and/or radially relative to the axis **26**. However, for ease of description, the acoustic panel **20** is described below with reference to the orientation of FIGS. **1** and **2** where a vertical direction extends radially relative to the axis **26**, a first lateral direction extends axially along the axis **26**, and a second lateral direction extends circumferentially about the axis **26**.

[0036] The acoustic panel **20** of FIGS. **1** and **2** includes a perforated face skin **34**, a solid (e.g., non-perforated) back skin **36** and a cellular core **38**. For ease of description, the face skin **34** is described below as an inner skin of the acoustic panel **20** and the back skin **36** is described below as an outer skin of the acoustic panel **20**. With such an arrangement, the acoustic panel **20** and its face skin **34** may form an outer peripheral boundary of at least a portion of the flowpath **24** within the aircraft propulsion system. It is contemplated, however, the face skin **34** may alternatively be the acoustic panel outer skin and the back skin **36** may alternatively be the acoustic panel inner skin with otherwise the same acoustic panel configuration of FIGS. **1** and **2**. With such an arrangement, the acoustic panel **20** and its face skin **34** may form an inner peripheral boundary of at least a portion of the flowpath **24** within the aircraft propulsion system. The present disclosure, of course, is not limited to the foregoing exemplary arrangements. The acoustic panel **20**, for example, may form a circumferential side boundary of the flowpath **24** and/or may otherwise be located with the aircraft propulsion system and/or the aircraft.

[0037] The face skin **34** of FIGS. **1** and **2** extends axially along and circumferentially about the axis **26**. The face skin **34** has a radial thickness **40**. This face skin thickness **40** extends radially between opposing exterior and interior sides **42** and **44** of the face skin **34**, where the face skin exterior side **42** is also the inner side **28** of the acoustic panel **20** of FIGS. **1** and **2**. The face skin thickness **40** may remain uniform (e.g., constant, the same) as the face skin **34** extends axially along and/or circumferentially about the axis **26**. Alternatively, the face skin thickness **40** may be varied as the face skin **34** extends axially along and/or circumferentially about the axis **26**, for example, to structurally reinforce one or more areas of the acoustic panel **20**, facilitate accommodations of an asymmetric material construction, etc.

[0038] Referring to FIG. **3**, the face skin **34** may be formed as composite skin. The face skin **34** of FIG. **3**, for example, includes one or more layers **46** of face skin material arranged in a stack **48** and bonded together to form a single monolithic member of the acoustic panel **20** (see FIGS. **1** and **2**). The face skin material may be a fiber-reinforced composite material. Each layer **46** of the face skin material, for example, may include a polymer matrix **50** and fiber reinforcement **52** embedded within the polymer matrix **50**. The polymer matrix **50** may be a thermoset material or a thermoplastic material. The polymer matrix **50**, for example, may be an epoxy, bismaleimides (BMI) resin, polyether ether ketone (PEEK) resin, or the like. The fiber reinforcement **52** may include fiberglass fibers, carbon fiber fibers, aramid (e.g., Kevlar®) fibers and/or the like. The fiber reinforcement **52** may be arranged as a (e.g., woven or unwoven) sheet of fibers and/or chopped fibers. The present disclosure, however, is not limited to such exemplary face skin materials. For example, in other embodiments, the face skin **34** may be formed from metal; e.g., sheet metal.

[0039] The face skin **34** includes a plurality of perforations **54**; e.g., apertures such as through-holes. The face skin perforations **54** are distributed axially and/or circumferentially along the face skin **34** and may (or may not) be equispaced from one another along the face skin **34**. Each of the face skin perforations **54** extends longitudinally along a centerline **56** of the respective face skin perforations **54** through the face skin **34** and its layers **46** from the face skin exterior side **42** to the face skin interior side **44**. The perforation centerline **56** of FIG. **3** is perpendicular to the face skin sides **42**, **44** at (e.g., on, adjacent or proximate) a location where the respective face skin perforation **54** pierces the face skin sides **42**, **44**. It is contemplated, however, the perforation centerline **56** may alternatively be (e.g., slightly) acutely angled relative to the face skin sides **42**, **44**.

[0040] The back skin **36** of FIGS. **1** and **2** extends axially along and circumferentially about the axis **26**. The back skin **36** has a radial thickness **58**. This back skin thickness **58** extends radially between opposing exterior and interior sides **60** and **62** of the back skin **36**, where the back skin exterior side **60** is also the outer side **30** of the acoustic panel **20** of FIGS. **1** and **2**. The back skin thickness **58** may remain uniform as the back skin **36** extends axially along and/or circumferentially about the axis **26**. Alternatively, the back skin thickness **58** may be varied as the back skin **36** extends axially along and/or circumferentially about the axis **26**, for example, to structurally reinforce one or more areas of the acoustic panel **20**, facilitate accommodations of an asymmetric material construction, etc. Referring again to FIGS. **1** and **2**, the back skin thickness **58** may be equal to or different (e.g., greater) than the face skin thickness **40**.

[0041] Referring to FIG. **4**, the back skin **36** may be formed as a composite skin. The back skin **36** of FIG. **4**, for example, includes one or more structural layers **64**A-D (generally referred to as "**64**") of structural back skin material and at least one fire-resistant layer **66** of fire-resistant back skin material arranged in a stack **68** and bonded together to form a single monolithic member of the acoustic panel **20** (see FIGS. **1** and **2**). The structural back skin material may be a fiber-reinforced composite material. Each structural layer **64** of the structural back skin material, for example, may include a polymer matrix **70** and fiber reinforcement **72** embedded within the polymer matrix **70**. The polymer matrix **70** may be a thermoset material or a thermoplastic material. The polymer matrix **70**, for example, may be an epoxy, bismaleimides (BMI) resin, polyether ether ketone (PEEK) resin, or the like. The fiber reinforcement **72** may include fiberglass fibers, carbon fiber fibers, aramid (e.g., Kevlar®) fibers and/or the like. The fiber reinforcement **72** may be arranged as a (e.g., woven or unwoven) sheet of fibers and/or chopped fibers. The present disclosure, however, is not limited to such exemplary first back skin materials. The fire-resistant back skin material may be or otherwise include an aerogel material **74** such as, but not limited to, Pyrogel XTE material produced by Aspen Aerogels of Northborough, Massachusetts, United States. The present disclosure, however, is not limited to such exemplary back skin materials.

[0042] The fire-resistant layer **66** of FIG. **4** is arranged at an intermediate location within the back skin stack **68**. The interior side structural layer **64**A, for example, may form the back skin interior side **62**. The exterior side structural layer **64**D may form the back skin exterior side **60**. The fire-resistant layer **66** of FIG. **4** is disposed (e.g., sandwiched) radially between the interior side structural layer **64**A and the exterior side structural layer **64**D. The interior side structural layer **64**A may be one of multiple of the structural layers **64**A-C radially between the fire-resistant layer **66** and the back skin interior side **62**, whereas the exterior side structural layer **64**D may (or may not) be the only one of the structural layers radially between the fire-resistant layer **66** and the back skin exterior side **60**. Here, the exterior side structural layer **64**D may form a protective covering over the fire-resistant layer **66** to protect the fire-resistant back skin material (e.g., the aerogel material **74**) from the from an environment **76** external to the acoustic panel **20** (see FIGS. **1** and **2**) and radially adjacent (or near) the back skin exterior side **60**. Note, while the structural back skin material may be selected for its structural properties and the fire-resistant back skin material may be selected for its fire resistant properties, the fire-resistant layer **66** may also add to a structural integrity of the back skin **36**. For example, similar to the structural layers **64**, the fire-resistant layer **66** is bonded to its radially adjacent structural layers **64**C and **64**D. This bonding may be through the polymer matrix **70** of the adjacent structural layers **64**C and **64**D and/or additional polymer matrix material may be added between the fire-resistant layer **66** and its radially adjacent structural layers **64**C and **64**D. Also, while the fire-resistant back skin material (e.g., the aerogel material **74**) may not be completely impregnated with the polymer matrix **70**, it is contemplated the polymer matrix **70** may migrate partially into open pores of the fire-resistant back skin material. The present disclosure, however, is not limited to the foregoing exemplary layered back skin construction. For example, in other embodiments, the exterior side structural layer **64**D may be omitted where the fire-resistant layer **66** is coated with a protective coating, etc. In still other embodiments, the layer **64**D may be a non-structural layer.

[0043] Referring to FIGS. **1** and **2**, the cellular core **38** is arranged and extends radially between the face skin **34** and the back skin **36**. One side of the cellular core **38**, for example, may be abutted radially against the face skin interior side **44**. Another side of the cellular core **38** may be abutted radially against the back skin interior side **62**. The cellular core **38** may also be connected to the face skin **34** and/or the back skin **36**. The cellular core **38**, for example, may be fused, adhered and/or otherwise bonded to the face skin **34** and/or the back skin **36**.

[0044] The cellular core **38** of FIGS. **1** and **2** extends axially along and circumferentially about axis **26**. The cellular core **38** has a radial thickness **78**. This core thickness **78** extends radially between and to the face skin **34** at its face skin interior side **44** and the back skin **36** at its back skin interior side **62**. The core thickness **78** may remain uniform as the cellular core **38** extends axially along and/or circumferentially about the axis **26**. Alternatively, the core thickness **78** may change; e.g., increase or decrease. The core thickness **78**, for example, may taper at or near one or more sides of the cellular core **38**. The core thickness **78** may be substantially larger than the face skin thickness **40** and/or the back skin thickness **58**. The core thickness **78**, for example, may be at least ten to forty times (10-40×), or more, larger than the face skin thickness **40** and/or the back skin thickness **58**. The cellular core **38** of the present disclosure, however, is not limited to such an exemplary dimensional relationship and may vary based on sound attenuation requirements, space requirements, etc.

[0045] The cellular core **38** of FIGS. **1** and **2** is configured with one or more internal core cavities **80** (e.g., open internal chambers, acoustic resonance chambers, etc.) radially between the face skin **34** and the back skin **36**. Referring to FIG. **5**, the cellular core **38** may be configured as a honeycomb core. The cellular core **38** of FIG. **5**, for example, includes a plurality of corrugated sidewalls **82**. These corrugated sidewalls **82** are arranged in a side-by-side array and are connected to one another such that each neighboring (e.g., adjacent) pair of the corrugated sidewalls **82** forms an array of the core cavities **80** laterally (e.g., circumferentially and/or axially) therebetween. The cellular core **38** and its corrugated sidewalls **82** may be constructed from or otherwise include a core material such as metal; e.g., sheet metal. The present disclosure, however, is not limited to such an exemplary cellular core construction nor material. For example, in other embodiments, the cellular core **38** and its corrugated sidewalls **82** may be constructed from or otherwise include a composite material similar to that of the face skin **34** and/or the back skin **36**.

[0046] Each core cavity **80** of FIGS. **1** and **2** extends radially within/through the cellular core **38** along a respective centerline **84** of the respective core cavity **80** between and to the face skin **34** at its face skin interior side **44** and the back skin **36** at its back skin interior side **62**. One or more or all of the core cavities **80** may thereby each overlap and be fluidly coupled with a respective set of one or more of the face skin perforations **54**. Referring to FIG. **5**, each of the core cavities **80** has a cross-sectional geometry (e.g., shape, size, etc.) when viewed in a reference plane; e.g., a plane perpendicular to the cavity centerline **84** of the respective core cavity **80**. This cavity cross-sectional geometry may have a polygonal shape; e.g., a hexagonal shape, a rectangular shape, a triangular shape, etc. The present disclosure, however, is not limited to foregoing exemplary cellular core configuration. For example, one or more or all of the core cavities **80** may alternatively each have a circular, elliptical or other non-polygonal cross-sectional geometry. Furthermore, various other types of honeycomb cores and, more generally, various other types of cellular cores for an acoustic panel are known in the art, and the present disclosure is not limited to any particular ones thereof.

[0047] The acoustic panel **20** of FIGS. **1** and **2** is configured as a single-degree of freedom (SDOF) acoustic panel. Each of the core cavities **80** of FIGS. **1** and **2**, for example, extends radially uninterrupted from the face skin **34** to the back skin **36**. With such an arrangement, the acoustic panel **20** may be tuned to attenuate, for example, a select frequency of sound, which tuning may be based on a radial height of each core cavity **80**/the core thickness **78**. The present disclosure, however, is not limited to single-degree of freedom acoustic panel applications. At least (or only) one perforated septum, for example, may be arranged in each of the core cavities **80** (or a subset of the core cavities **80**) to configure the acoustic panel **20** as a multi-degree of freedom (e.g., a double-degree of freedom) acoustic panel. Various types and configurations of acoustic panel septums are known in the art, and the present disclosure is not limited to any particular ones thereof.

[0048] During operation of the acoustic panel **20** of FIGS. **1** and **2**, sound waves may enter a respective core cavity **80** through the respective face skin perforation(s) **54**. These sound waves may travel through the core cavity **80** and reflect against the back skin **36**. The reflected sound waves may travel back through the core cavity **80** and exit the acoustic panel **20** through the respective face skin perforation(s) **54**, where those reflected sound waves may be out of phase from and destructively interfere with incoming soundwaves. Of course, the sound waves may also or alternatively reflect against one or more other elements of the acoustic panel **20** which may further influence sound attenuation.

[0049] Various components **86** of the aircraft propulsion system (schematically shown in FIGS. **1** and **2**) may be arranged in the environment **76** adjacent (or near) the panel outer side **30**. Examples of these propulsion system components **86** include, but are not limited to, an accessory gearbox, a working fluid pump (e.g., a fuel pump, a lubricant pump, a coolant pump, a hydraulic pump, etc.), a working fluid line (e.g., a fuel line, a lubricant line, a coolant line, a hydraulic line, etc.), an electrical device (e.g., a generator, an engine controller, etc.), and the like. In an unlikely event of a fire located at and/or a malfunction of one of the propulsion system components **86**, the acoustic panel **20** and its panel outer side **30** may be subject to relatively high temperatures and/or flames. Under such conditions, the fire-resistant layer **66** and its fire-resistant back skin material (e.g., the aerogel material **74**) may protect a majority of the back skin **36** as well as the face skin **34** and the cellular core **38** (e.g., when constructed from the composite material) from the high temperatures which may otherwise degrade those acoustic panel elements; e.g., decompose the polymer matrix within the acoustic panel elements. In addition, by integrating the fire-resistant back skin material into the back skin **36**, the acoustic panel **20** may be configured in the aircraft propulsion system without other discrete fire-resistant elements (e.g., a fire blanket) covering the acoustic panel **20**. This in turn may reduce costs, manufacturing steps and/or space requirements of the acoustic panel **20** as compared to an acoustic panel which may be paired with an additional, discrete fire-resistant element such as a fire blanket. Encapsulating the thermal protection may also reduce maintenance compared to a fire blanket which may be susceptible to damage since it is laid over a component to be protected.

[0050] In some embodiments, referring to FIG. **6**, the fire-resistant layer **66** may extend along an entirety of a lateral (e.g., axial and/or circumferential) width of the acoustic panel **20**. The fire-resistant layer **66** of FIG. **6**, for example, extends laterally (e.g., axially and/or circumferentially) to each respective edge **88** of the acoustic panel **20**. In other embodiments, referring to FIGS. **7** and **8**, the fire-resistant layer **66** may extend partially along the lateral width of the acoustic panel **20**.

Each edge **90** of the fire-resistant layer **66** of FIG. **7**, for example, is laterally (e.g., axially and/or circumferentially) spaced from a respective acoustic panel edge **88**.

[0051] In some embodiments, referring to FIGS. **6** and **7**, the fire-resistant layer **66** may extend laterally (e.g., axially and/or circumferentially) along and thereby overlap and cover an entirety of the cellular core **38**. The fire-resistant layer **66** of FIG. **6**, for example, extends laterally (e.g., axially and/or circumferentially) beyond (or alternatively to in FIG. **7**) each respective edge **92** of the cellular core **38**. The fire-resistant layer **66** thereby laterally (e.g., axially and/or circumferentially) overlaps and covers each of the core cavities **80** (e.g., open volumes within the cellular core **38** as described above) as well as one or more plugged cavities **94** (e.g., cavities partially or completely filled by material such as the polymer matrix) in the cellular core **38**. Note, a portion of the back skin **36** extending laterally along the plugged cavities **94** is subject to reduced pressure loads during aircraft propulsion system operation since pressurized air flowing within the flowpath **24** cannot directly contact the back skin **36** at its back skin interior side **62** through the plugged cavities **94**. In other embodiments, referring to FIG. **8**, the fire-resistant layer **66** may extend partially laterally along the cellular core **38**. The fire-resistant layer **66** of FIG. **8**, however, may still laterally (e.g., axially and/or circumferentially) overlap and cover each of the core cavities **80**. Here, the fire-resistant layer **66** partially laterally overlaps each structural layer **64** (collectively labeled in FIGS. **6-9**) while at least some or all of the structural layers **64** may each completely laterally overlap and project laterally beyond the fire-resistant layer **66**.

[0052] In some embodiments, referring to FIG. **9**, the back skin thickness **58** may change as the back skin **36** extends laterally (e.g., axially and/or circumferentially) along the cellular core **38**. The back skin thickness **58** of FIG. **9**, for example, decreases as the back skin **36** extends laterally (e.g., axially and/or circumferentially) away from each edge **90** of the fire-resistant layer **66** into an intermediate region **96** of the back skin **36** and its fire-resistant layer **66**. The back skin thickness **58**, for example, may be tailored to account for structural reinforcement provided to the back skin **36** by inclusion of the fire-resistant layer **66**.

[0053] In some embodiments, referring to FIGS. **6-9**, a side portion of the cellular core **38** may taper to a point where the face skin **34** and the back skin **36** come together. Cavities in this tapered portion of the cellular core **38** may be the plugged cavities **94**. However, it is contemplated at least some of the cavities in the tapered portion of the cellular core **38** may alternatively be the open core cavities **80**.

[0054] FIG. **10** illustrates an example of the aircraft propulsion system with which one or more of the acoustic panels **20** may be configured. This aircraft propulsion system includes a turbofan gas turbine engine **98**. The gas turbine engine **98** of FIG. **10** extends axially along an axial centerline **100** (e.g., the axis **26**) between an upstream airflow inlet **102** and a downstream combustion products exhaust **104**. The gas turbine engine **98** includes a fan section **106**, a compressor section **107**, a combustor section **108** and a turbine section **109**. The turbine section **109** includes a high pressure turbine (HPT) section **109**A and a low pressure turbine (LPT) section **109**B, which LPT section **109**B may also be referred to as a power turbine (PT) section.

[0055] The engine sections **106-109**B are arranged within the engine housing **112**. This engine housing **112** includes an inner housing structure **114** and an outer housing structure **116**. The inner housing structure **114** may house one or more of the engine sections **107-109**B; e.g., a core of the gas turbine engine **98**. The outer housing structure **116** may house at least the fan section **106**. The inner and the outer housing structures **114** and **116** of FIG. **10** also form a bypass duct. The inner and/or the outer housing structure **116** may each include one or more of the acoustic panels **20**.

[0056] Each of the engine sections **106**, **107**, **109**A and **109**B includes a respective bladed rotor **118-121**. Each of these bladed rotors **118-121** includes a plurality of rotor blades arranged circumferentially around and connected to one or more respective rotor disks. The rotor blades, for example, may be formed integral with or mechanically fastened, welded, brazed, adhered and/or otherwise attached to the respective rotor disk(s).

[0057] The fan rotor **118** is connected to and driven by the LPT rotor **121** through a low speed shaft **124**. The compressor rotor **119** is connected to and driven by the HPT rotor **120** through a high speed shaft **126**. The shafts **124** and **126** are rotatably supported by a plurality of bearings (not shown). Each of these bearings is connected to the engine housing **112** by at least one stationary structure.

[0058] During operation, air enters the gas turbine engine **98** through the airflow inlet **102**. This air is directed through the fan section **106** and into a core flowpath **128** and a bypass flowpath **130** (e.g., the flowpath **24**). The core flowpath **128** extends sequentially through the engine sections **107-109**B. The air within the core flowpath **128** may be referred to as “core air”. The bypass flowpath **130** extends through the bypass duct, which bypasses the engine core. The air within the bypass flowpath **130** may be referred to as “bypass air”.

[0059] The core air is compressed by the compressor rotor **119** and directed into a combustion chamber **132** of a combustor **134** in the combustor section **108**. Fuel is injected into the combustion chamber **132** and mixed with the compressed core air to provide a fuel-air mixture. This fuel air mixture is ignited and combustion products thereof flow through and sequentially cause the HPT rotor **120** and the LPT rotor **121** to rotate. The rotation of the HPT rotor **120** drives rotation of the compressor rotor **119** and, thus, compression of the air received from a core airflow inlet. The rotation of the LPT rotor **121** drives rotation of the fan rotor **118**, which propels the bypass air through and out of the bypass flowpath **130**. The propulsion of the bypass air may account for a majority of thrust generated by the gas turbine engine **98**.

[0060] The acoustic panel **20** may be included in various gas turbine engines other than the one described above. The acoustic panel **20**, for example, may be included in a geared gas turbine engine where a geartrain connects one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the acoustic panel **20** may be included in a gas turbine engine configured without a geartrain; e.g., a direct drive gas turbine engine. The acoustic panel **20** may be included in a gas turbine engine configured with a single spool, with two spools (e.g., see FIG. **10**), or with more than two spools. The gas turbine engine may be configured as a turbofan engine, a turbojet engine, a turboprop engine, a turboshaft engine, a propfan engine, a pusher fan engine or any other type of gas turbine engine. The gas turbine engine may alternatively be configured as an auxiliary power unit (APU). The present disclosure therefore is not limited to any particular types or configurations of gas turbine engines. Furthermore, while the acoustic panel **20** is described above with respect to various aircraft applications, the acoustic panel **20** of the present application may alternatively be used for non-aircraft applications.

[0061] While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An apparatus for an aircraft, comprising: an acoustic panel including a face skin, a back skin and a cellular core disposed between and connected to the face skin and the back skin; the back skin laterally overlapping the face skin and comprising an aerogel layer; and the cellular core comprising a plurality of cavities that extend through the cellular core from the face skin to the back skin, and each of the plurality of cavities fluidly coupled with one or more perforations in the face skin.

2. The apparatus of claim 1, wherein the aerogel layer laterally overlaps each of the plurality of cavities.

3. The apparatus of claim 1, wherein the back skin further comprises a fiber-reinforced layer bonded to and laterally overlapping the aerogel layer.

4. The apparatus of claim 3, wherein the fiber-reinforced layer completely laterally overlaps the aerogel layer; and the aerogel layer partially laterally overlaps the fiber-reinforced layer.

5. The apparatus of claim 3, wherein the fiber-reinforced layer is disposed between the cellular core and the aerogel layer.

6. The apparatus of claim 5, wherein the fiber-reinforced layer is a first fiber-reinforced layer; the back skin further comprises a second fiber-reinforced layer bonded to and laterally overlapping the aerogel layer; and the aerogel layer is disposed between the first fiber-reinforced layer and the second fiber-reinforced layer.

7. The apparatus of claim 3, wherein the aerogel layer is disposed between the fiber-reinforced layer and the cellular core.

8. The apparatus of claim 3, wherein the fiber-reinforced layer comprises fiber reinforcement embedded within a polymer matrix.

9. The apparatus of claim 8, wherein the fiber reinforcement comprises fiberglass.

10. The apparatus of claim 8, wherein the fiber reinforcement comprises carbon fiber.

11. The apparatus of claim 1, wherein a thickness of the back skin decreases as the back skin extends away from an edge of the aerogel layer into an intermediate region of the aerogel layer.

12. The apparatus of claim 1, wherein the face skin comprises fiber reinforcement embedded within a polymer matrix.

13. The apparatus of claim 1, wherein the cellular core comprises metal.

14. The apparatus of claim 1, wherein the cellular core comprises a honeycomb core.

15. The apparatus of claim 1, wherein the acoustic panel extends axially along and circumferentially about an axis; and the back skin is disposed radially outboard of the face skin.

16. An apparatus for an aircraft, comprising: a first skin comprising a first fiber-reinforced composite; a second skin laterally overlapping the first skin, the second skin including a second fiber-reinforced composite and a fire-resistant material bonded to and laterally overlapping the second fiber-reinforced composite; and a metal core disposed between and bonded to the first skin and the second skin, the metal core comprising a plurality of cavities that extend through the metal core from the first skin to the second skin.

17. The apparatus of claim 16, wherein each of the plurality of cavities is fluidly coupled with one or more apertures extending through the first skin.

18. The apparatus of claim 16, wherein the fire-resistant material comprises an aerogel material.

19. An apparatus for an aircraft, comprising: a first skin; a second skin laterally overlapping the first skin and comprising an aerogel layer; and a cellular core disposed between and bonded to the first skin and the second skin, the cellular core comprising a plurality of cavities that extend through the cellular core from the first skin to the second skin; wherein a thickness of the second skin changed as the aerogel layer extends laterally along the plurality of cavities.

20. The apparatus of claim 19, further comprising an acoustic panel including the first skin, the second skin and the cellular core.